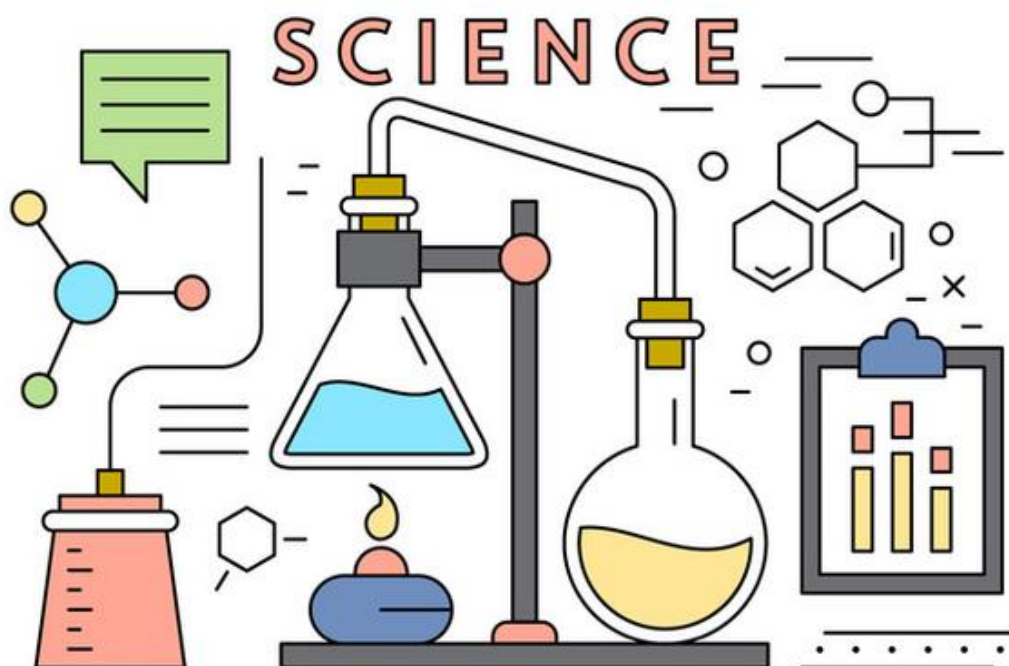


Fanling Kau Yan College

Integrated Science

Chapter 1 Introducing Science

Learning Guide (Teacher's Version)



「自從造天地以來，神的永能和神性是明明可知的，雖是眼不能見，但藉著所造之物，就可以曉得，叫人無可推諉。」(羅馬書 1:20)

For from the first making of the world, those things of God which the eye is unable to see, that is, his eternal power and existence, are fully made clear, he having given the knowledge of them through the things which he has made, so that men have no reason for wrongdoing. (Romans 1:20)

Name: _____ **Class:** _____ ()

Contents

Assessment for learning :

After studying this chapter, please evaluate whether you can achieve the learning objective, put a ✓ on the suitable box.

Learning Guides	Sections	In this lesson, I have learnt (1 is the least, 5 is the most)	Remarks
1	1.3B Laboratory safety rules	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
2	1.3C Hazard warning labels	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
3	1.3A First visit to the laboratory (Equipment)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
	1.3E What to do in case of fire?	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
4	1.4B Basic experimental skills (Lighting up a Bunsen burner)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
5	1.4B Basic experimental skills (Comparing Bunsen flames)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
6	1.2A Basic steps of a scientific investigation	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
	1.2D Methods for doing scientific investigations (Fair Test)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	

Common question words

Question words	Chinese meaning
Arrange ... in the correct order	以正確次序排列
Ascending	由小至大 / 由少至多
Descending	由大至小 / 由多至少
Can....? Should we....? / Does....?	可否...? / 能否...? (需回答 yes or no)
Circle	圈出
Classify	分類
Complete (the following table)	完成 (下表)
Do you agree?	你是否同意? (需回答 yes or no)
Draw	畫出
Explain	解釋
Finish	完成
Give	給
How	如何
How do we test for the presence of (a gas)? (State the test and the expected observation)	如何測試(某氣體)的存在? (需回答用甚麼物質作測試 及 實驗結果)
Label	標示
List	列出
Match	配對
Name	命名
Plot the graph	繪製一幅圖表
Read	閱讀
State	陳述
Suggest	建議
Underline	下橫線
What	甚麼
When	何時
Where	哪裡
Which	哪一個
Why	為甚麼
Write down	寫下

You must follow the laboratory rules (實驗室規則) when working in the laboratory.

Do	Don't
1. Follow your <u>teacher's instruction</u>. 2. Wear safety goggles when heating or mixing <u>substances</u>. 3. Tie back long hair and <u>fasten</u> school ties when doing experiments. 4. Keep the <u>laboratory</u> clean and tidy. 5. Report all <u>accidents</u> to the teacher at once. 6. Keep an eye on your <u>experiment</u>.	1. Enter the laboratory without teacher's <u>permission</u>. 2. Do experiments without teacher's permission. 3. Eat, drink, play or run about in the laboratory. 4. Point the mouth of a test tube towards anyone when heating. 5. Leave a lighted Bunsen burner <u>unattended</u>.

Index

★ = Questions are of elementary levels / answers can be retrieved directly from textbook.

⚡ = Questions are of more advanced levels and require deductions or higher-order thinking.

Learning Guide 1

1.3B Laboratory safety rules

Objectives:**Knowledge:** (1) Recognize the potential dangers and accidents in the laboratory.

(2) State the meaning of some common hazard warning labels.

Skill: Prevent potential dangers by following laboratory safety rules.**Attitude:** Build up the concept of laboratory safety.★Q1. **Keywords: Laboratory Safety**(I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).(II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.(III) Review the keywords and prepare for **dictation**.

Dictionary
(with pronunciation)



Laboratory safety	Diagrams
(a) laboratory (noun) (實驗室)	
(b) potential danger (noun phrase) (潛在危險)	
(c) safety precaution (noun phrase) (安全守則)	
(d) safety goggles (noun phrase) (安全眼鏡)	
(e) safety gloves (noun phrase) (安全手套)	

Hints: 潛在危險、安全守則、實驗室、安全眼鏡、安全手套

Preparation Questions

(Read textbook P.23–28 and answer the following questions)

(Check the Chinese meaning of the words **highlighted** before lesson)★Q2. We usually do **experiments** in a laboratory.★Q3. There are potential dangers when working in the **laboratory**.⚡Q4. The following picture shows **SIX** **potential dangers**. Write down the potential dangers on the lines **provided** below. The first one has been done for you.

- (1) e.g. The students' hands may be cut by the broken glass pieces.
- (2) The student may take in something dirty (e.g. chemicals or germs) on the pen.
- (3) The students may hit others when running in the laboratory.
- (4) The hot liquid may spill out and hurt other people.
- (5) The beaker may drop easily.
- (6) The student may get an electric shock.

★Q5. In order to **avoid accidents**, we should **follow** the laboratory safety rules.

★Q6. Give **THREE** “**DOs**” and **THREE** “**DON'Ts**” in the laboratory.

DO's	DON'Ts
Follow your teacher's instructions. (or other acceptable answers.)	Do not enter the laboratory without your teachers' permission. (or other acceptable answers.)
Keep all exits and passages clear. (or other acceptable answers.)	Do not do experiments without your teacher's permission. (or other acceptable answers.)
Wear <u>safety goggles</u> when heating or mixing substances.	Do not touch any chemicals or hot objects with <u>bare hands</u> .

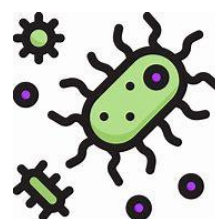
Reflection Questions

★Q1. Key point

- ◆ We should follow the laboratory safety rules in the laboratory to avoid accidents.

Extension Task

Read the following passage and answer the questions below.



In the past, people believed that diseases were caused by bad air. But scientists like Louis Pasteur and Robert Koch discovered tiny organisms called **germs** that can make us sick. They showed that many diseases are caused by these germs, like **bacteria** and **viruses**. This discovery helped us develop better ways to prevent and treat illnesses.

From this passage, we can tell that scientific knowledge is subject to change (*is subject to change / will never change*) as new evidence (證據) becomes available.

Learning Guide 2

1.3C Hazard warning labels

Preparation Questions

(Read textbook P.29–33 and answer the following questions)

(Check the Chinese meaning of the words **highlighted** before lesson)

★Q1. **Keywords: Hazard Warning Labels**

- (I) Underline the **vowels** (a, e, i, o, u) and break up the words into **syllables** (only for the first word).
 (II) Check the dictionary/textbook and write the Chinese meaning of the words in the brackets provided.
 (III) Review the keywords and prepare for dictation.

Dictionary
(with pronunciation)



Hazard warning labels	Diagrams
(a) corrosive (adj.) (腐蝕性)	
(b) flammable (adj.) (易燃)	
(c) toxic (adj.) (有毒)	
(d) explosive (adj.) (爆炸性)	

Hints: 腐蝕性、有毒、爆炸性、易燃

★Q2. There are many **chemicals** in the laboratory. They can be hazardous in **different** ways. They may be **harmful**, **irritant**, corrosive, flammable, toxic or explosive. **Hazard warning labels** are put on the **containers** of these chemicals.

⚡Q3. When using each type of dangerous chemical, what **safety precaution** should we take? For questions (a) and (b), put a tick (✓) in the **appropriate** boxes. For questions (c) and (d), write down the answers.

(a)


☐ Keep them away from fire and heat.

☒ Wear safety goggles and gloves.

☐ Use them in a fume cupboard.

(b)


☒ Keep them away from fire and heat.

☒ Wear safety goggles and gloves.

☐ Use them in a fume cupboard.

(c)


 1. Wear safety goggles and gloves.

 2. Use them in a fume cupboard.

(d)


 1. Keep them away from fire and heat.

 2. Wear safety goggles and gloves.

★Q4. **Key point**

- ◆ We should read the hazard warning labels on the **containers** of dangerous chemicals carefully, and must **handle** them with **caution**.



Experiment: The magic liquid

Teacher's Demonstration

Chemicals	Materials / Apparatus	Safety precaution
♦ water ♦ ethanol	♦ 100 mL beaker × 1	 Safety goggles must be worn



Q5. Why was there a fire in the beaker of water?

There was a layer of flammable liquid (ethanol) on top of the water.

The fire goes off when all the ethanol is used up.

The magic liquid



Reflection Questions

(TB P.30 Checkpoint 1.3B)

Checkpoint 1.3B

1. If an experiment involves toxic gases, do it in a fume cupboard.
2. If a chemical is irritant, wear a pair of safety gloves to protect your hands and a pair of safety goggles to protect your eyes.



3. What should you do if some chemical spills on your skin? (Hint: TB P.33)

Slowly wash the affected area with running water.

4. After this lesson, how does your view towards science lesson has **CHANGED**?

I used to think	Now I think
(Accept any reasonable answers)	(Accept any reasonable answers)

Learning Guide 3**1.3A First visit to the laboratory (Equipment)****1.3E What to do in case of fire?****Objectives:**

Knowledge: (1) Name the basic equipment found in a laboratory.
 (2) Recognize the fire triangle and various ways of putting out a fire.

Skill: State the necessary conditions for a fire to start.

Attitude: Build up the concept of laboratory safety.

- (I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).
 (II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.
 (III) Review the keywords and prepare for **dictation**.

Dictionary

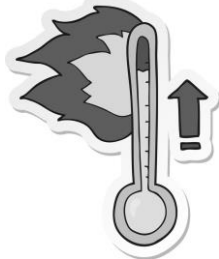
(with pronunciation)

★Q1. **Keywords (I): Equipment**

Equipment (noun) (儀器)	Diagrams
(a) fume cupboard (noun phrase) (通風櫃)	
(b) fire extinguisher (noun phrase) (滅火筒)	
(c) gas tap (noun phrase) (煤氣掣)	
(d) water tap and sink (noun phrase and noun) (水龍頭和水槽)	

Hints: 儀器、水龍頭和水槽、煤氣掣、通風櫃、滅火筒

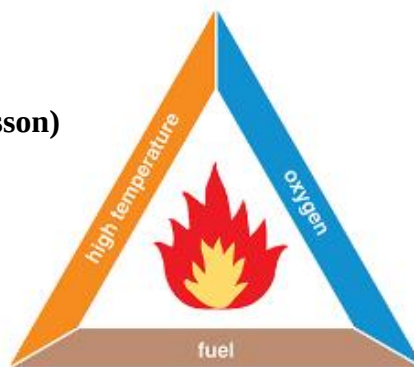
★Q2. Keywords (II): Fire triangle

Fire Triangle (<i>noun</i>) (火三角)	Diagrams
(a) fuel (<i>noun</i>) (燃料)	
(b) oxygen (<i>noun</i>) (氧氣)	
(c) high temperature (<i>noun phrase</i>) (高溫)	

Hints: 火三角、高溫、氧氣、燃料

Preparation Questions

(Read textbook P.34–35 and answer the following questions)

(Check the Chinese meaning of the words **highlighted** before lesson)★Q3. The figure on the right shows a fire triangle.**Keypoints:**

★Q4. For a fire to start, three **conditions** are needed: fuel, oxygen and high temperature. These three conditions form the fire triangle.

★Q5. To put out a fire, we can remove any one or more of the conditions in the fire triangle.



Experiment: The burning candle

Step 1	Light up a candle.	
Step 2	Cover the candle with a beaker.	
Step 3	Observe the candle flame.	

Results and analysis:

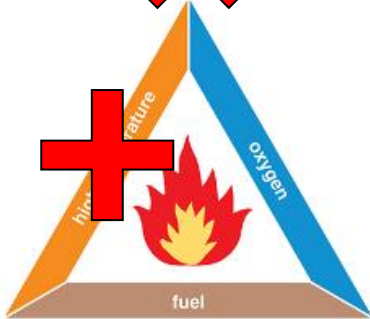
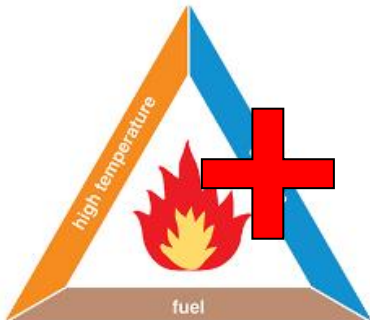
Q6. What happens to the candle flame after some time?

The candle flame goes off.

Q7. What does this experiment tell you?

We can put off a fire by removing the oxygen supply.

Q8. Below shows how people put out fires in different cases. Cross out (put a “✕”) which conditions in the fire triangle is removed in each case.

Cases	Pictures	Cross out which conditions in the fire triangle is removed:
Case 1 Turning off a gas stove.		
Case 2 Pouring water on a barbecue fire.		
Case 3 Covering a fire with a blanket.		

★ Q9. Key point

- ◆ For a fire to start, fuel, oxygen and a high temperature are needed. These three conditions form the fire triangle.
- ◆ To put out a fire, we can remove any one or more of the conditions in the fire triangle.



Q10. More to know:

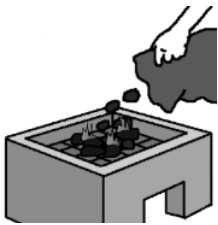
Why do the following methods can **put off** a fire? Complete the table below to learn more.

Methods of putting out a fire	How the method works	Condition(s) in the fire triangle removed
Turning off a gas stove	This can <u>cut off</u> the fuel supply to the fire.	fuel
Pouring water on a barbecue fire.	Water can <u>lower</u> the temperature of the surroundings .	high temperature
Covering a fire with a blanket.	The blanket can <u>cut off</u> the oxygen supply to the fire.	oxygen



Q11. More to know:

Why do the following methods can make the fire **stronger**? Complete the table below to learn more.

Methods of making a fire stronger	How the method works	Condition(s) in the fire triangle added
Add more charcoal to the barbeque fire 	This can <u>supply</u> the fuel to the fire.	fuel
Fanning the fire with a piece of newspaper 	Fanning can <u>supply</u> oxygen to the fire.	oxygen

Reflection Questions

(TB P.35 Checkpoint 1.3C)

Checkpoint 1.3C

1. The three common types of fire-fighting equipment found in the laboratory are (a) fuel, (b) oxygen and (c) high temperature.
2. The following pictures show different ways of putting out a fire. Which condition(s) in the fire triangle is/are removed in each case?



fuel



high temperature (or oxygen)

Extension Task

Read the following passage and answer the questions below.

Tutorial on Portable Fire Extinguishers

Here are the steps of using a **portable** fire extinguisher:



All portable fire extinguishers inside train **compartments** are the powder version which is applicable to different **flammables** (noun). In case of fire, keep **calm**, move to a safer place and **seek help**.

1. What are the **FOUR** steps of using a fire extinguisher?

Step 1: Pull the pin.

Step 2: Aim at the base of the fire.

Step 3: Squeeze the handle.

Step 4: Sweep from side to side.

2. (a) If oil catches fire, why should we **NOT** to use water to put out the fire?

The water will boil at once and the hot oil will splash out.

- (b) What should we do if oil catches fire? Use a fire extinguisher / fire sand.

Learning Guide 4**1.4B Basic experimental skills (Lighting up a Bunsen burner)****Objectives:****Knowledge:** Name different parts of a Bunsen burner.**Skill:** Light up a Bunsen burner.**Attitude:** Build up the concept of laboratory safety.

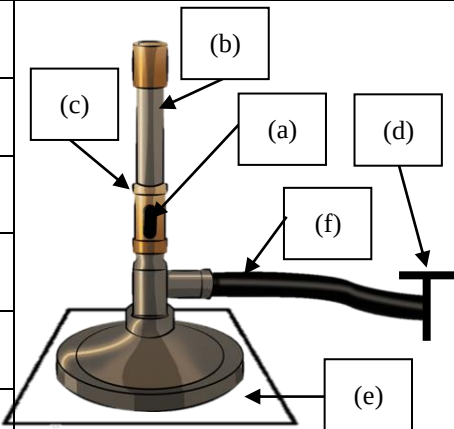
- (I) Underline the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).
- (II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.
- (III) Review the keywords and prepare for **dictation**.

Dictionary
(with pronunciation)



★Q1. **Keywords: Structure of a Bunsen burner**

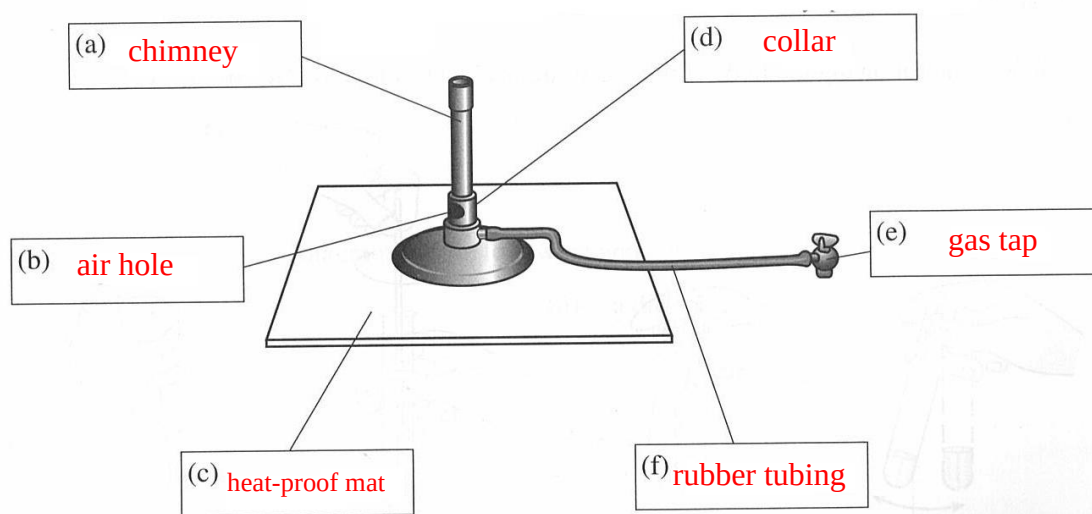
Bunsen burner (noun) (本生燈)		
(a) air hole	(noun phrase)	(氣孔)
(b) chimney	(noun)	(煙囪)
(c) collar	(noun)	(空氣調節器)
(d) gas tap	(noun phrase)	(煤氣掣)
(e) heat-proof mat	(noun phrase)	(防熱墊)
(f) rubber tubing	(noun phrase)	(膠喉)



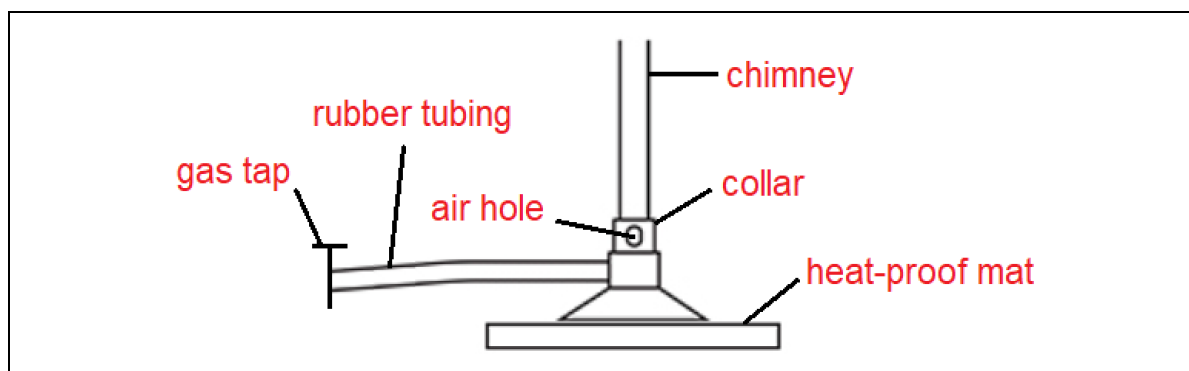
Hints: 本生燈、氣孔、煤氣掣、防熱墊、煙囪、空氣調節器、膠喉

Preparation Questions

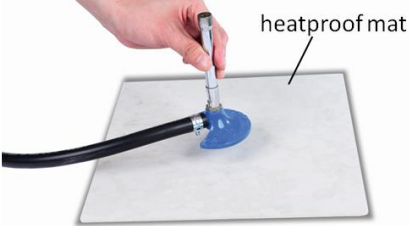
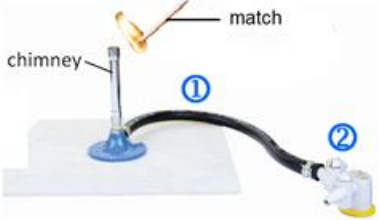
(Read textbook P.41–46 and answer the following questions)

(Check the Chinese meaning of the words **highlighted** before lesson)★Q2. Label the **structure** of a Bunsen burner.★Q3. What is the **function** of a Bunsen burner?For heating.★Q4. Below are **TWO** **safety precautions** / safety rules when using a Bunsen burner. Fill in the blanks.*e.g.* Make sure the laboratory is **well-ventilated**.

- (1) Wear safety goggles.
- (2) Put a heat-proof mat under the Bunsen burner.

**[Skill Corner – Teacher's **Demonstration**]**Q5. Draw and **label** the structure of a Bunsen burner.

★Q6. Fill in the blanks of the **procedures** of lighting up a Bunsen burner.

Pictures	Procedures
	Safety precaution: Wear <u>safety goggles</u> before lighting up the Bunsen burner.
	1. Check if the <u>rubber tubing</u> is tightly connected to the <u>gas tap</u>.
	2. Put a <u>heat-proof mat</u> under the Bunsen burner.
	3. Turn the collar to <u>close the air hole</u>.
	4. Light up a match. Put the tip of the match over the <u>chimney</u> first, and then turn on the gas tap.
	5. Slowly turn the <u>collar</u> to open the <u>air hole</u> halfway.
 (Turning off the Bunsen flame)	6. Close the <u>air hole</u> and turn off the <u>gas tap</u>.

Teacher's demonstration:

Watch how the teacher lights the Bunsen burner and turns it off.

**Experiment Time**

- (1) Take turns to light up the Bunsen burner and turn it off.
- (2) **Assess** the skills of classmates in lighting the Bunsen burner. (*Circle the marks*)

Checking		Marks		
1.	He / She wears safety goggles.	2	1	0
2.	He / She makes sure that the rubber tubing is tightly connected .	2	1	0
3.	He / She puts the Bunsen burner on a heat-proof mat.	2	1	0
4.	He / She closes the air hole before lighting up.	2	1	0
5.	He / She opens the air hole halfway after lighting up.	2	1	0
Total marks		_____ / 10		

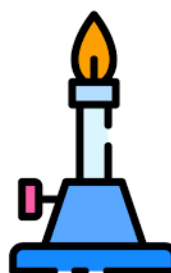
★Q7. **Key point**

- ◆ In the laboratory, we often use a **Bunsen burner** for heating.



- Q8. (TB P.44 **Caution**) If the air hole is opened before lighting up the Bunsen burner, the gas may burn inside the chimney, and you will hear a strange noise. This is called **striking back**. If **striking back** happens, turn off the **gas tap** at once and tell your teacher.

Video: Striking back of
Bunsen burner



Reflection Questions

1. Below are the steps involved in lighting a Bunsen burner. **Arrange** them in the correct **order** by writing letters in the boxes.

(A) Put the tip of a lighted match over the chimney, and then turn on the gas tap.

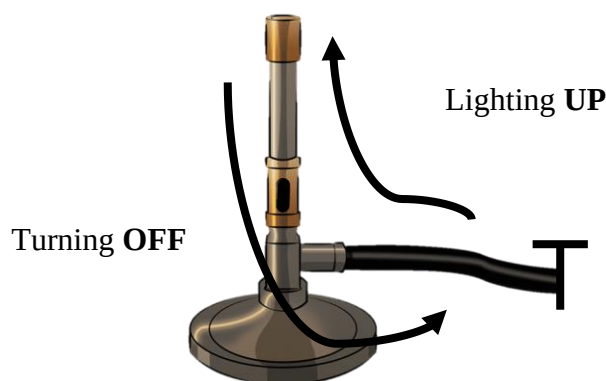
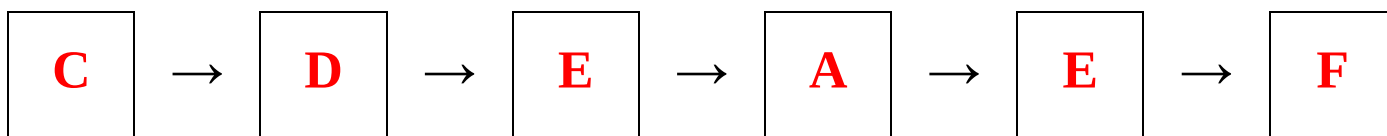
(B) Turn the collar to make the air hole half-opened.

(C) Check if the rubber tubing is tightly connected to the Bunsen burner.

(D) Put a heat-proof mat under the Bunsen burner.

(E) Turn the collar to close the air hole.

(F) Close the air hole and turn off the gas tap.



2. List **TWO** **safety precautions** when using a Bunsen burner.

(1) Wear safety goggles.

(2) Put a heat-proof mat under the Bunsen burner.

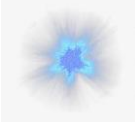
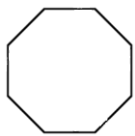
Learning Guide 5**1.4B Basic experimental skills (Comparing Bunsen flames)****Objectives:****Knowledge:** State the differences between luminous and non-luminous Bunsen flames.**Skill:** Produce luminous and non-luminous Bunsen flames by adjusting the air hole.**Attitude:** Build up the concept of laboratory safety.

- (I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).
- (II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.
- (III) Review the keywords and prepare for **dictation**.

Dictionary
(with pronunciation)



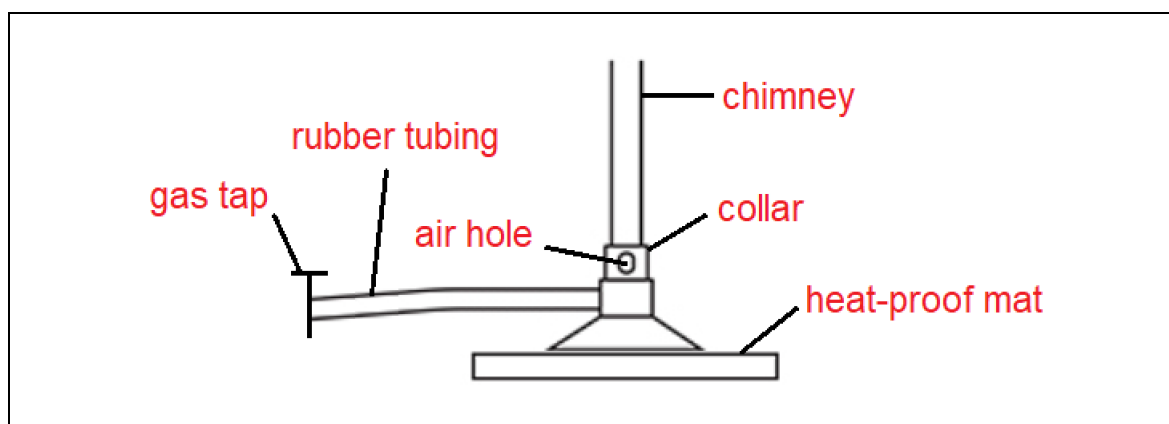
★Q1. **Keywords: Comparing different Bunsen flames**

Comparing different Bunsen flames	Diagrams
(a) luminous (adj.) (光亮的)	
(b) non-luminous (adj.) (不光亮的)	
(c) regular (adj.) (規則的)	
(d) irregular (adj.) (不規則的)	
(e) quiet (adj.) (安靜的)	
(f) noisy (adj.) (嘈雜的)	

Hints: 規則的、不規則的、光亮的、不光亮的、安靜的、嘈雜的

Preparation Questions

(Read textbook P.45–46, 48, 61–62 and answer the following questions)

(Check the Chinese meaning of the words **highlighted** before lesson)★Q2. Draw the **sectional diagram** of a Bunsen burner. Label different parts.**Experiment 1: Comparing different Bunsen flames**

- (1) Light the Bunsen burner.
- (2) **Observe** the Bunsen flames when the air hole is closed and half opened. Record your observations.
- (3) Turn off the gas tap after use.

Q3. Finish the table below.

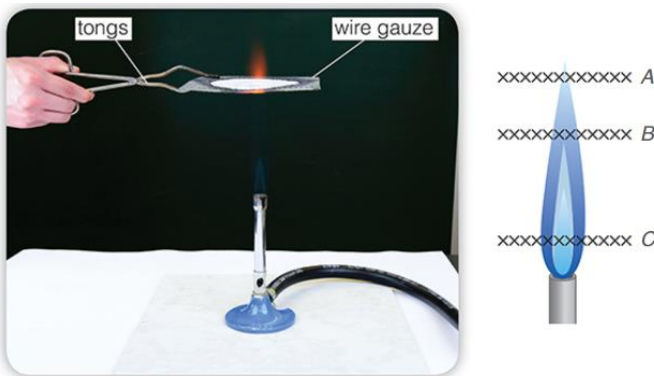
Results

	Air hole closed	Air hole half opened
Shape of flame (Draw the flame observed.)		
Colour of flame	yellow	blue
Brightness of flame	<u>luminous*</u> / non-luminous*	luminous / <u>non-luminous</u>
Noise level	<u>quiet</u> / noisy	quiet / <u>noisy</u>



Experiment 2: Knowing more about Bunsen flames (Teacher's Demonstration)



Step 1	Light the Bunsen burner. Half open the air hole halfway to get a <u>non-luminous</u> flame.
Step 2	Hold a piece of <u>wire gauze</u> with a pair of <u>tongs</u> at position A of a non-luminous flame as shown. Start the <u>stopwatch</u> . 
Step 3	When the wire gauze turns <u>red hot</u> , stop the stopwatch. Record the <u>time</u> needed (in seconds) for the wire gauze to turn red hot.
Step 4	Let the wire gauze cool down <u>completely</u> on a <u>heat-proof mat</u> .
Step 5	Take two other pieces of wire gauze. <u>Repeat</u> steps 2 to 4 for positions B and C respectively.

Results

Position	A	B	C
Time needed for the wire gauze to turn red hot	(Answers depend on the wire gauze and the Bunsen burner used.)		

Analysis

- In which position does the wire gauze turn red hot in the shortest time? B
- What part of the Bunsen flame is the hottest? B

Q4. **Conclusion:** the middle (*upper / middle / lower*) part of the Bunsen flame is the hottest.

Reflection Questions

★Q5. Key point

- When the air hole is closed, the flame is luminous, irregular in shape and quiet.
- When the air hole is half opened, the flame is non-luminous, regular in shape and noisy.
- The middle part of the Bunsen flame is the hottest.

Extension Task

Read the following passage and answer the questions below.



TODAY IN CHEMISTRY HISTORY

31ST MARCH – BUNSEN BURNER DAY



ROBERT BUNSEN

BORN

31 March 1811

DIED

16 August 1899



1855



1860



1861

Robert Bunsen is best known for developing the Bunsen burner, as well as discovering caesium and rubidium.

THE FLAME

The coolest flame (safety flame) is yellow/orange, due to incandescent soot particles. The hotter, 'roaring' flame is blue.

THE BARREL

The barrel raises the height of the flame, allowing it to be used more easily for heating.

THE FUEL

Methane is usually the gas used from gas taps to fuel Bunsen burners. Both propane and butane can also be used as fuels.



THE BUNSEN BURNER

THE COLLAR

The collar allows the flow of air into the burner to be adjusted. A blue, roaring flame is produced when it is completely open; a safety flame is produced when it is closed.

THE AIR HOLE & GAS JET

The air hole allows air into the burner (adjustable with the collar). The air is drawn in by the rush of gas ejected from the gas jet.



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1. Name the **scientist** who **invented** the Bunsen burner. Robert Bunsen
2. What is the year of **invention** of Bunsen burner? 1855
3. What is the other name for “chimney”? Barrel

Learning Guide 6**1.2A Basic steps of a scientific investigation****1.2D Methods for doing scientific investigations (Fair test)****Objectives:****Knowledge:** State the variable to be changed, variable to be kept constant and variable to be compared.**Skill:** Construct a table to identify the important variables in a fair test.**Attitude:** Build up the concept of fair testing.

- (I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).
- (II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.
- (III) Review the keywords and prepare for **dictation**.

Dictionary
(with pronunciation)

★ Q1a. **Keywords (I): Scientific investigation**

Scientific investigation	(noun phrase)	(科學探究)	Diagrams
(a) observation	(noun)	(觀察)	
(b) hypothesis	(noun)	(假說)	
(c) result	(noun)	(結果)	
(d) conclusion	(noun)	(結論)	

Hints: 觀察、假說、結論、結果、科學探究

★Q1b. Keywords (II): Fair test

Fair test (noun phrase) (公平測試)			Diagrams
(a) variable	(noun)	(變項)	
(b) changed	(adj.)	(改變)	
(c) constant	(adj.)	(不變)	
(d) compared	(verb)	(比較)	

Hints: 公平測試、改變、比較、不變、變項

Preparation Questions

(Read textbook P.14–15 and answer the following questions)

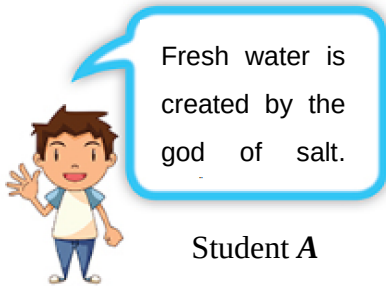
(Check the Chinese meaning of the words **highlighted** before lesson)★Q2. There are **FIVE** basic steps for scientific investigation:

- (1) Making observations and asking questions.
- (2) Doing **background** research.
- (3) **Proposing** a hypothesis.
- (4) **Designing** and **conducting** experiments.
- (5) **Analyzing** results and drawing a conclusion.

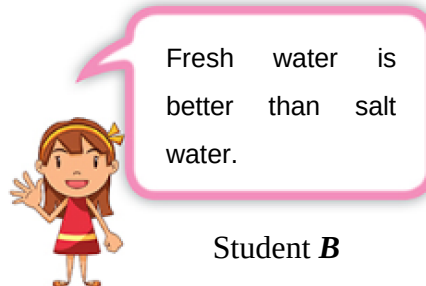
★Q3. A hypothesis is a reasonable answer to the question. It must be **testable**.



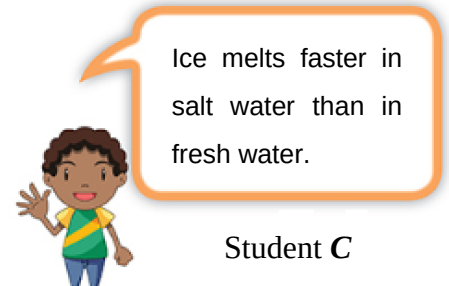
Q4. Below shows three hypothesis suggested from three students, **A**, **B** and **C**.



Student **A**



Student **B**



Student **C**

Which student gives the best hypothesis? **Explain** your answer.

Student **C** gives the best hypothesis. Because it can be tested.

Student **A** gives a hypothesis which cannot be tested.

Student **B** gives a hypothesis which is subjective.

Language Support:

We use the following word phrase patterns to describe the variables in fair test.

e.g. Volume of water
 (**attributes**) of (**noun**)

Q5. Make use of following words to form three **variables** that may appear in fair test.

attributes: size, **volume**, **speed**, **height**, **weight**, temperature, **length**, **shape**

noun: water, Tom, ball, bottle, salt, apple, car

Variable 1: Size of ball

Variable 2: Height of Tom

Variable 3: Temperature of water (Accept other reasonable answers)



Q6. The student **conducts** an experiment to **investigate** the **factors** that may **affect** the time for melting a ball of ice in water. Suggest some of the factors you found may be useful.

(1) *e.g. temperature of water*

(2) Size of the ice ball

(3) Shape of the ice ball

(4) Temperature of the air

The factors listed here are called variables.



Q7. To find out how other variables affect the time, we can **vary** one variable each time while keeping **others the same**. In this way, we have done a **fair test**.



Q8. The following table shows three experiments.

(a) Circle the **difference(s)** in the experimental set-up.

(b) **Decide** whether the test is a fair test or not.

	Experiment A	Experiment B	Experiment C
Circle the difference in the set up	ice ball 1 L fresh water	ice ball 1 L fresh water	ice ball 1 L fresh water
How many difference(s)?	2	2	1
Is this experiment a fair test?	This (is / <u>is not</u>) a fair test	This (is / <u>is not</u>) a fair test	This (<u>is</u> / is not) a fair test



Q9. Finally, the variables in the experiment C can be tabulated as follows:

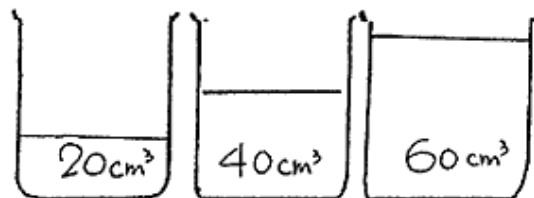
Variable to be changed	Variables to be kept constant	Variable to be compared
types of water	- sizes of ice - shapes of ice - volumes of water etc	time of ice melting



[Skill corner – Teacher’s Demonstration]

- Q10. *Case study 1:* Simon puts some **salt** into a glass of water. He wants to know whether the volume of water will **affect** the amount of salt dissolved. He thinks that more salt will **dissolve** in a larger volume of water. To test his hypothesis, he carries out a scientific investigation.

If there is more water, I can dissolve more salt!



- (a) What is the problem to be investigated?

Whether the volume of water will affect the amount of salt dissolved.



For finding the problem, always look for the “W” word in the question.

- (b) What is Simon’s hypothesis?

More salt will dissolve in a larger volume of water.



What did Simon think?

- (c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
volume of water	- temperature of water - size of the cup - size of the salt	maximum amount of salt that can dissolve in water

Hints:	volume of water	temperature of water	size of the cup
	maximum amount of salt that can dissolve in water		size of the salt



Q11. Case study 2: Susan wants to know which **brand** of washing powder (brand **A** or brand **B**) has a better **cleaning effect**. She thinks that washing powder of brand **A** has a higher ability to remove **soya sauce stain** than that of brand **B**. To test her hypothesis, she carries out a scientific investigation.



For finding the problem,
always look for the “W”
word in the question.

(a) What is the problem to be investigated?

Which brand of washing powder has a better cleaning effect.

(b) What is Susan’s hypothesis?

What did Susan think?

Washing powder of brand **A** has a higher ability to remove soya sauce stain than that of brand **B**.

(c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
brand of washing powder	- temperature of water - size of the stain - amount of washing powder - volume of water - method of washing	time of cleaning the stain

Hints:	temperature of water	size of the stain	volume of water
	brand of washing powder	time of cleaning	method of washing



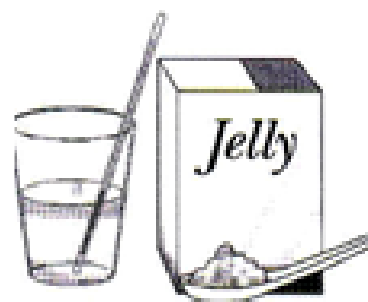
Experiment: Dissolving jelly powder with water of different temperatures.

Preparations:

Chemicals	Materials / Apparatus	Safety precaution
◆ 80 °C water ◆ 25 °C water ◆ Jelly powder (1 bag)	◆ 100 mL beaker × 2 ◆ Glass rod × 2 ◆ Spatula × 2 ◆ Stopwatch	Be careful of hot water



Q12. *Case study 3:* You are now going to do a scientific investigation on whether hot water dissolves jelly powder faster than cold water. You think that using hot (*hot / cold*) water will dissolve jelly powder faster than using cold (*hot / cold*) water.



(a) What is the problem to be investigated?

Whether hot water dissolves more jelly powder than cold water.

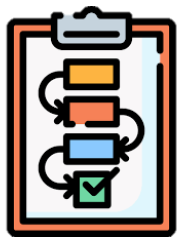
(b) What is your hypothesis?

Hot water will dissolve jelly powder faster than using cold water.

(c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
• temperature of water	• size of the cup • time of stirring • brand of jelly powder	• time needed to dissolve all the jelly powder

Procedure



1. **Label** one **beaker** as “**hot water**” and the other as “**cold water**”.
2. **Measure** about 80 mL of 80 °C water and pour it into the “hot water” beaker.
3. Measure about 80 mL of 25 °C water and pour it into the “cold water” beaker.



hot water



cold water

4. Use the **spatula** to measure an **equal amount** of jelly powder from the bag and **transfer** it to the beaker labeled “hot water”. Start the **stopwatch** after adding all the powder.



5. **Stir** the mixture using the **glass rod** until the jelly powder is **completely dissolved**. Stop the stopwatch and record the time taken to dissolve all the powder.



6. Repeat **Steps 4–5** using the beaker labelled “cold water”.

Results and Discussion

(d) Write down the time needed to dissolve all the jelly powder in hot and cold water respectively.

Temperature	25 °C (<i>cold water</i>)	80 °C (<i>hot water</i>)
Time (in seconds) needed to dissolve all the jelly powder	<i>(Depends on the experiment)</i>	<i>(Depends on the experiment)</i>

(e) Which kind of water, hot water or cold water, has a shorter **dissolving** time? Hot water



Conclusion (*what did this experiment tell you?*)

From the experiment, hot water dissolves jelly powder faster (*faster / slower*) than cold water.

13. After this activity, how does your **view** towards scientific investigation has **CHANGED**?

I used to think	Now I think
<i>(Accept any reasonable answers)</i>	<i>(Accept any reasonable answers)</i>

Reflection Questions

1. Mary wants to find out whether **plants** **grow** better in water or **soil**. She thinks that plants grow better in soil. To test her hypothesis, she carries out a scientific investigation.



- (a) What is the problem to be investigated?

Whether plants grow better in water or soil.

- (b) What is Mary's hypothesis?

Plants grow better in soil.

- (c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
environment of growth	(initial) size of the plant - temperature of the room - amount of sunlight	height of the plant

Hints: temperature of the room **initial** size of the plant **environment** of **growth**
height of the plant after 10 days **amount** of **sunlight**